

Towards Trustworthy Respiratory Audio Intelligence: An Open-Set, Physician-Benchmarked Evaluation Framework for Lightweight CNN and Transformer Backbones

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Abstract

Respiratory sound classification usually relies on a single public benchmark and one headline score. This project investigates the hidden details behind that metric. Breathing cycles are segmented and converted to log-scaled spectrograms. A lightweight convolutional neural network and three pretrained vision transformers are trained on the official patient disjoint partition. Evaluation covers standard classification metrics, parameter count, and training time.

The CNN achieves the highest challenge score of 0.5602. The top transformer requires 12 times the parameters but scores 3 points lower. Another transformer simply predicts the majority class, gaining high raw accuracy only because normal sounds make up 59 percent of the corpus. Spectrogram masking improves the CNN by 4 points but offers no benefit to the transformers. Grouping the classes into a binary abnormal versus normal task increases the score by 7 points. Most model errors happen when distinguishing crackles from wheezes, a task where human physician agreement is under 0.40.

The best model is evaluated further using a twelve-row ablation study, noise floor testing, and explainable artificial intelligence. Testing on the 47-patient partition shows major differences between cycle-level and patient-level analyses. The overall performance ceiling comes from the dataset labels and sample size rather than the network architecture.

Index Terms

convolutional neural network, data augmentation, evaluation protocol, explainable artificial intelligence, ICBHI 2017, label reliability, pre-registration, reproducibility, respiratory sound classification, transfer learning, vision transformer.

I. INTRODUCTION

Physicians usually reach for a stethoscope first to examine patients with respiratory conditions. Heartbeats and background noises often hide important breathing sounds like crackles and wheezes. Diagnoses vary a lot from one doctor to another because of these listening challenges. The ICBHI 2017 Respiratory Sound Database [1] serves as a standard tool to help automate this detection process using machine learning. This project selects this problem because the benchmark is mature. The objective is not to create a completely new architecture. The goal is to ask different questions about the testing process. The project investigates whether the scores reported by the field on this dataset actually mean what people assume they mean.

A. Literature Review

Yu *et al.* [2] have reviewed 135 technical publications built on this database. They separated the tasks into two. To recognise the patient's pathological state, where many systems exceed 98 percent accuracy. To categorise the adventitious sound in a single cycle. Severe class imbalance makes the second task much more difficult. They suggest using the data surrounding the audio more effectively and creating models that are easier to understand. Recent work is driven in that direction. Since around 2021, progress has come more from improved pre-training than from improved architecture. On the official challenge metric [3], convolutional systems pretrained on ImageNet [4] reach a plateau around 0.56. Transformers that have been pretrained on AudioSet [5] have scores that are five to nine points higher [6], [7], [8]. Suma *et al.* [9] use a single-input, multi-output network over mel-frequency cepstral coefficients to simultaneously address both tasks. Four backbones are compared. MobileNet does the best, scoring 91 percent on diseases and 74 percent on sounds. Which partition those numbers are on is never mentioned. That is the initial indication of the issue this paper addresses. Gains of about one point are reported by two 2025 fusion systems [10], [11], a scale at which the following measurement questions become significant.

Bae *et al.* [6] fine-tuned an Audio Spectrogram Transformer [12] with a patch-mixing contrastive objective and achieved 0.6237. Niizumi *et al.* [13] froze the audio foundation model entirely, trained only a linear probe, and still achieved 0.5938. That is above several fully fine-tuned convolutional systems. Most of the useful information sits in the pretrained representation, not the fine-tuning, provided the pretraining was on audio. Moummad and Farrugia [14] achieved 0.5755 using a small CNN and contrastive pretraining on metadata. Nguyen and Pernkopf [15] add that the transfer from ImageNet to spectrograms is fragile. It needs regularisation. Kim *et al.* [16] pushed the metadata idea furthest. They came to a decision that we had to face head-on. They assessed the sources of domain shift, including age, sex, chest position, and recording equipment. A supervised contrastive objective was guided by the most informative pairings, and the main limiting factor for models trained on this

corpus was found to be recording-condition bias. The results of our search for that axis in the actual data are presented in Section II-A.

To examine the reasons behind a prediction, concept bottleneck models [17] filter it using human-defined concepts. This is appealing since wheezes and crackles have physical definitions. This was our original plan. Karim *et al.* [18] combined vision transformers with OpenMax rejection. We reimplemented that as our open-set baseline. Knowledge distillation with a teaching assistant [19] gave us the compression arm. Yang *et al.* [20] has done the vocabulary separating out-of-distribution detection from open-set recognition. Kejriwal *et al.* [21] has done the staged open-world protocol. Cho and Lee [22] are the only group found to have performed open-set recognition on respiratory audio. They reported an improvement over a semi-supervised baseline, not an absolute score. Cruz *et al.* [23] surveyed what the open-world field still has not solved. They group the open problems into three areas. The theory of novelty, the design of agents, and their evaluation. The third area aligns with us. They name inconsistent evaluation metrics as an unresolved issue.

Tzeng *et al.* [24] had seven senior physicians blindly annotate a quarter of the ICBHI test set. On clean audio, they reached an ICBHI score of 47.77 percent, with a sensitivity of 23.23 percent, and a mean self-rated confidence of 2.88 out of 5. Melbye *et al.* [25] had twelve physicians classify twenty reference recordings. The agreement fell below $\kappa = 0.40$ for detailed descriptions. It rose to 0.62 and 0.59 when crackles and wheezes were collapsed to a simple presence. Huang *et al.* [26] report a larger clinical archive with agreement on wheezes being very high ($\kappa = 0.948$) and agreement on crackle being only moderate ($\kappa = 0.516$). The three studies disagree on the level. They do agree on the shape, that crackle labels are far less reliable than wheeze labels. As the description becomes more detailed, reliability declines dramatically. Bikku *et al.* [27] combined a recurrent-convolutional encoder over mel-spectrograms from two corpora. They associate wheezes with obstructive disease and crackles with pneumonia. They report 94.0 percent accuracy under five-fold cross-validation and use gradient, Shapley, and local surrogate methods to explain the result. The explanations align with the same cycle labels that the three studies characterized earlier. The partition is not the official one. The number cannot be read against the systems in the first block. Both observations shaped how we report our own interpretability results in Section III-L.

B. Gaps in the Related Work

Reading the two groups side by side exposed four gaps we could act on.

The reported score is treated as a characteristic of the model. Papers from the first group report their composite numbers and compare them with those of other papers. A few of them state the circulating definition they used. The description of the partition is often vague or non-existent. The multi-task study does not name any. Five-fold cross-validation is reported in the convolutional-recurrent study, which can't be compared to the official split figures next to it. We could not find an ICBHI paper that ablates its evaluation protocol the way it ablates its architecture. Our own audit shows the protocol is worth more than any architectural choice we made.

The official split is assumed to be independent of the patient, but it is not. The published file assigns recordings, not patients. Two patients have recordings on both sides. That is a validity fault that no paper we read mentions.

The reference standard is never characterized. The first group validates models against ICBHI's cycle labels. The second shows that those labels are reproduced by trained physicians at only 47.77 percent, with agreement (κ) well below 0.40 for fine distinctions. A detector scored against a reference standard of unknown reliability produces a number of unknown meanings.

We could not find a paper that set a validity threshold in advance, ran a test against it, and reported a failure. Without that, a reader cannot tell how many attempts a reported success survived.

C. Our Work

We started with an open-world multi-task design: a shared spectrogram backbone, one head for the four sound-event classes, one head for disease, and a bottleneck of physics-derived acoustic concepts in between. Before building the bottleneck we fixed a threshold. The extractors had to reproduce ICBHI's own crackle and wheeze labels at an area under the curve of at least 0.65, with an interval that excludes chance, read once per run on the test partition, and we budgeted two runs. Crackle reached 0.5506 and 0.5580. Wheeze reached 0.5729 and 0.5340. The rule said stop, and we stopped.

Looking for a reason sent us into our own evaluation code, where we found five faults. Section IV lists them. Correcting them moved our headline number from 0.7077 to 0.5602 on the same model and the same code. That gap of 0.1475 is larger than ImageNet pretraining, larger than augmentation, and larger than the spread between our best and worst backbone. What we deliver is a measurement rather than a mechanism: a corrected protocol, the baselines it produces, and the checks that let somebody else catch each fault in their own pipeline.

We rebuilt the model side under that protocol. One ImageNet-pretrained convolutional network and three ImageNet-pretrained vision transformers [28], [29], [30] were trained on identical inputs with identical schedules, each once clean and once with SpecAugment [31]. A fourth transformer, an Audio Spectrogram Transformer [12] pretrained on AudioSet, was trained before the audit and stored no prediction table, so its challenge-metric score cannot be recovered. Section III-B reports that gap instead of repairing it, because it is our own argument landing on the one model the literature says should have won.

The best model, MobileNetV2 [32] with SpecAugment, went through a twelve-row ablation, a three-seed noise floor, paired patient-level tests, and gradient and occlusion attribution [33]. Every transformer lost to the convolutional baseline and the largest collapsed to the majority class. We report that instead of tuning until it stops being true. The attribution argues against our own score: with a wheeze present the model attends to the wheeze band less than when it is absent, by 0.0534 at $p = 0.001$, and that measurement is what put the zero-padding row into the ablation.

The contributions of this work are as follows.

- We quantify the evaluation protocol as an ablatable pipeline component on ICBHI 2017, reporting a five-fault protocol ablation that moves an unchanged MobileNetV2 checkpoint from 0.5602 to 0.7077 (Δ 0.1475), exceeding the combined effect of ImageNet pretraining (0.0603), spectrogram masking (0.0402) and backbone choice.
- We benchmark the reference standard itself with a blinded single-physician study (132 clips, 6 calibration exemplars, 12 hidden duplicates), reproducing a published seven-physician crackle sensitivity at 23.1 percent against 23.23 percent, and measure a paired +0.169 AUROC advantage for a frozen audio probe over that physician on identical clips (wheeze $p = 0.046$; crackle undecided at $p = 0.087$).
- We identify checkpoint selection as a first-class pipeline component, measuring a 0.0752 to 0.1015 penalty across three seeds for minimum-loss selection against challenge-score selection, and scope the measurement to a shared monitoring partition so that only the difference between the two rules is claimed.
- We document a patient leak in the published 60/40 split (patients 156 and 218 on both sides), price its repair at 0.0039, and separate validity from inflation by re-testing ten ablation rows at both units of analysis (7 of 10 significant per cycle, 1 of 10 per patient, 6 verdict flips over 2,636 cycles from 47 patients).
- We report corrected-protocol baselines for four backbones trained under one schedule (MobileNetV2, Swin-T, DeiT-S and ViT-B/16, each clean and masked), gated by a three-seed admissibility floor of 0.0141 that marks 3 of 11 ablation deltas unresolved. Every ImageNet-pretrained transformer loses to the 2.23 M-parameter convolutional baseline, and ViT-B/16 (85.8 M) collapses to the majority class.
- We pre-register a concept-validity gate ($\text{AUROC} \geq 0.65$ with a DeLong interval excluding chance, two-run budget), report its failure at 0.5580 on crackle and 0.5340 on wheeze, and retract our own label-noise explanation using a pre-registered frozen-AudioSet probe that reaches 0.7115 and 0.7621 on the identical cycles.
- We quantify attribution against the score rather than illustrating it, measuring a -0.0534 band-pointing deficit when wheeze is present ($p = 0.001$) and a 0.098 mean attribution correlation across identical tiled copies of one cycle, the second of which motivated a zero-padding ablation row costing 0.0311.

II. PROPOSED SYSTEM

All numerical results are saved for Section III. The numbers here are characteristics of the data. The entire system is depicted in Fig. 1 below. It shows the locations of each of the five faults listed in Section IV.

A. Dataset

We use the ICBHI 2017 Respiratory Sound Database [1] from its public release. It includes 920 recordings from 126 patients, annotated per respiratory cycle with a start time, an end time, a crackle flag, and a wheeze flag. Combining the two flags gives us the four sound-event classes used throughout. They are labeled as Normal, Crackle, Wheeze, and Both across 6,898 annotated cycles, with a median length of 2.42 s. We standardize that to an 8.0 s network input. Recordings were captured with four stethoscopes at several chest locations and at mixed sample rates. We resample it all to 16 kHz mono.

Two structural facts about the corpus shaped the project and are worth stating before any results. Each patient carries a single diagnosis label. Those labels are extremely imbalanced: COPD alone covers 64 of the 126 patients, 83.3 percent of all cycles. That is why the disease side of our original two-head design was eventually dropped from the headline. Only four patients (112, 158, 218, and 226) had multiple device records. Patient identification and device identity are almost perfectly confounded. It is important to note that recording-condition bias is the primary limiting factor for models trained here, according to Kim *et al.* [16]. They demonstrate the importance of an axis that this corpus cannot measure internally. Separating the device from the patient would require that patients be recorded on several devices, but there are only four. It is a reason to take their cross-dataset approach seriously, not a reason to doubt it.

Table I presents the three partitions used in this paper, along with the class composition of each. The published split file assigns recordings, not patients. Patients 156 and 218 have recordings on both sides. Our corrected partition moves every recording of a leaking patient into the training side. That is the conservative direction. The test side is guaranteed to contain no patient the model has trained on. The repair costs 12 of 381 test recordings. It moves the ratio from 58.6/41.4 to 59.9/40.1, marginally closer to the nominal 60/40 than the published file itself. All new results in this paper are on the corrected partition. The other two partitions appear only where a comparison demands it, always by name.

On the corrected test partition, both classes hold 86 cycles against 1,560 Normal. This shows how severe the imbalance is. That is why every model here trains with an inverse-frequency weighted loss (5). It is also why accuracy is a misleading headline on this corpus: always predicting Normal scores 0.5918 accuracy on that partition, higher than the accuracy of our best model.

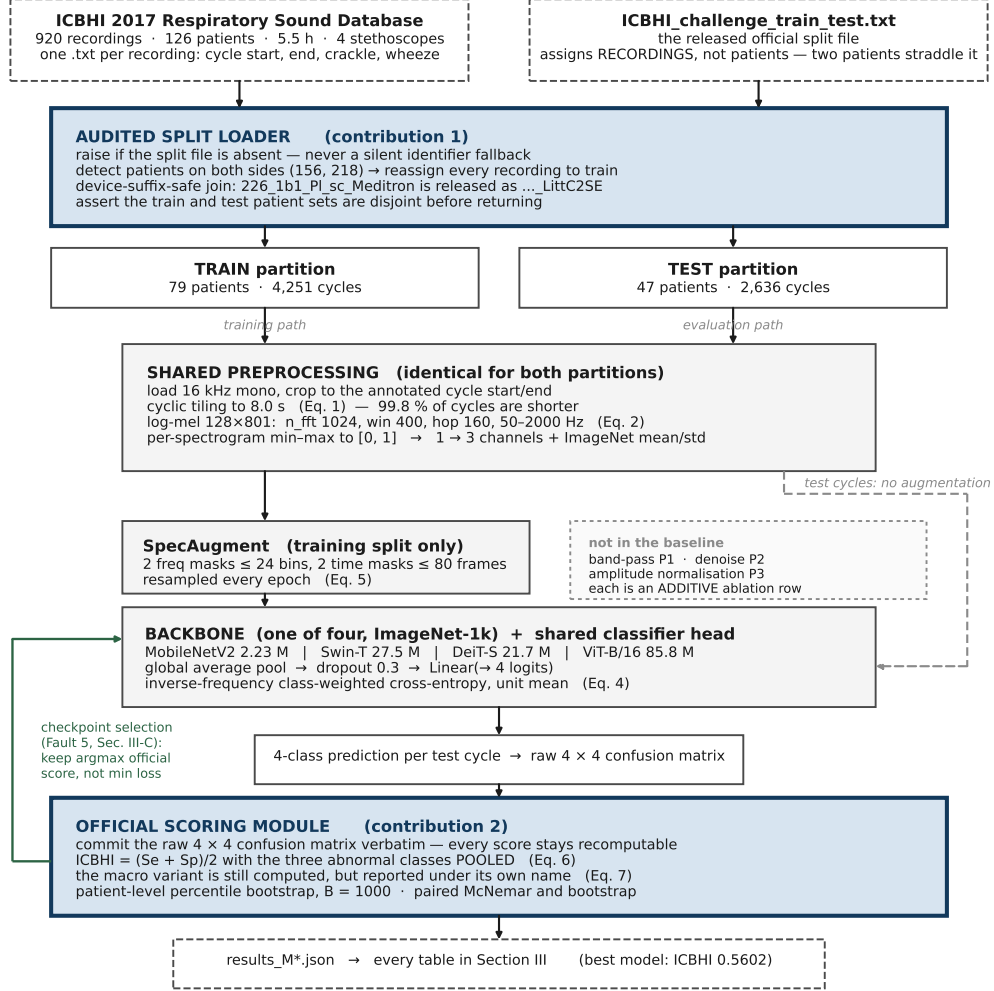


Fig. 1. Flowchart of the complete system. Audio is segmented into respiratory cycles, standardized in length, converted to a log-mel spectrogram, and classified into four sound-event classes.

TABLE I
CLASS COMPOSITION OF THE THREE PARTITIONS USED IN THIS PAPER, COMPUTED FROM THE ICBHI ANNOTATION FILES. MASKING ADDS NO SAMPLES, SO THE AUGMENTED AND CLEAN TRAINING TOTALS ARE EQUAL.

Partition	Subset	Normal	Crackle	Wheeze	Both	Total	Aug. total	Patients	Rec.
Whole corpus	—	3,642	1,864	886	506	6,898	—	126	920
Corrected (ours)	Train	2,082	1,247	513	420	4,262	4,262	79	551
	Test	1,560	617	373	86	2,636	—	47	369
Published, verbatim	Train	2,063	1,215	501	363	4,142	4,142	77	539
	Test	1,579	649	385	143	2,756	—	49	381
Identifier fallback	Train	3,387	1,700	848	471	6,406	6,406	115	—
	Test	255	164	38	35	492	—	11	—

The corrected partition reassigns the two leaking patients to training; the published partition is the split file verbatim; the identifier-fallback partition is not a design but a fault (Section IV). Two earlier backbone runs use a fourth policy that instead *drops* the leaking patients' training recordings and leaves the published test set byte-identical, so their test row is the published test row. Pooling the abnormal classes, the corrected test partition holds 703 crackle-positive and 459 wheeze-positive cycles, the denominators of every concept result in Section III-I. One recording named 226_1b1_P1_sc_Meditron in the split file is released as 226_1b1_P1_sc_LittC2SE, so a stem join drops it and its 11 cycles; it is a training recording, which is why our pipeline indexes 4,251 training cycles where the annotations hold 4,262.

B. Dataset Preprocessing

The chain is implemented once. Every stage is switchable. That lets the ablation in Section III-G change one thing at a time. The execution order is:

- 1) Load and resample to 16 kHz mono.
- 2) Optionally, band-pass filter with a zero-phase fourth-order Butterworth between 50 and 2000 Hz.
- 3) Segment into cycles at the annotated start and end times.
- 4) Standardise every cycle to 8.0 s by cyclic tiling, Eq. (1).
- 5) Optionally normalize each cycle to unit peak amplitude.
- 6) Optionally denoise by spectral gating at the 25th percentile, with an over-subtraction factor of two.
- 7) Compute a log-mel spectrogram, Eq. (2), with 128 mel bands, a 1024-point FFT, hop 160, and window 400, over 50 to 2000 Hz.
- 8) Rescale each spectrogram to the unit interval, Eq. (3).
- 9) Replicate to three channels for the pretrained backbones.
- 10) Encode the label, Eq. (4).
- 11) Handle imbalance in the loss, Eq. (5).
- 12) Apply spectrogram masking, Eq. (6), to the training split only.

Three of those stages are off in the reference configuration and are done deliberately. They are band-pass filtering, denoising, and per-cycle amplitude normalization. Turning any of them on in the reference would silently move the baseline that every other ablation delta is measured against. Instead, they are switched on one at a time, as rows P1, P2, and P3 of Table VIII. Their deltas read in the opposite direction to every other row: a positive delta is a recommendation to adopt the stage.

The median cycle is 2.42 s, against an 8.0 s input. Almost every cycle has to be extended. We tile rather than pad with silence. For a cycle x of length L , and a target of $N = T f_s$ samples:

$$\tilde{x}[n] = x[n \bmod L], \quad n = 0, \dots, N - 1, \quad (1)$$

A longer cycle is cropped from the start. Tiling is common practice here. It repeats the same acoustic event two or three times within a single input. It leaves a discontinuity at each seam. Row P4 tests the alternative for reasons provided in Section III-L. The tiled waveform is converted to a log-mel spectrogram. $X[m, k]$ is the short-time Fourier transform. $H_j[k]$ is the j -th mel filter:

$$S[j, m] = 10 \log_{10} \left(\frac{\sum_k H_j[k] |X[m, k]|^2}{\max_{j', m'} \sum_k H_{j'}[k] |X[m', k]|^2} \right). \quad (2)$$

The maximum is per spectrogram, not global. That is itself a normalisation. It is why the explicit rescaling that follows is nearly free in the ablation. That rescaling is a min-max map to the unit interval:

$$\tilde{S} = \frac{S - \min S}{\max S - \min S + \epsilon}, \quad \epsilon = 10^{-8}. \quad (3)$$

Labels come from the two annotation flags: c is the crackle flag, w is the wheeze flag, both in $\{0, 1\}$:

$$y = c + 2w \in \{0, 1, 2, 3\} \equiv \{\text{Normal, Crackle, Wheeze, Both}\}. \quad (4)$$

Instead of resampling, imbalance is addressed in the loss. This maintains the identical training distribution across each model and ablation row. The raw inverse-frequency weight for class c is renormalised to the unit mean for N_c of N training cycles and $C = 4$:

$$w_c = \frac{N}{CN_c} \bigg/ \frac{1}{C} \sum_{c'=1}^C \frac{N}{CN_{c'}}, \quad \mathcal{L} = -\frac{1}{B} \sum_{i=1}^B w_{y_i} \log p_{i, y_i}. \quad (5)$$

Without renormalisation, the loss scale moves with the class distribution. Ablation rows that change the training set would no longer be comparable to the reference. SpecAugment [31] is applied only to training spectrograms, using the same seed as the paired clean run. The two runs differ in exactly one thing:

$$\tilde{S}[j, m] \leftarrow 0 \quad \text{for } j \in [f_0, f_0 + f), \quad f \sim \mathcal{U}[0, 24], \quad \text{and } m \in [t_0, t_0 + t), \quad t \sim \mathcal{U}[0, 80], \quad (6)$$

Two frequency masks, two time masks, per spectrogram.

C. Applied Models

Table II lists every model, with its owner. The four backbones trained under the corrected protocol share the same spectrograms, loss, partition, batch size of 16, cap of 40 epochs, and seed 42. Anything else would turn the comparison into a comparison of tuning effort. Only the classifier head is replaced. The two families do differ in optimizer settings. We state that rather than smoothing it over. The convolutional model employs Adam with a single learning rate of 5×10^{-4} , dropout of 0.3, and gradient clipping of 5.0. Vision transformers use AdamW with a learning rate of 10^{-4} on the backbone and 10^{-3} on the head, which is the conventional way to fine-tune a pretrained transformer. Both use cosine annealing and weight decay of 10^{-4} . The ablation harness reuses the convolutional settings unchanged.

TABLE II

EVERY MODEL IN THIS PAPER, WITH THE GROUP MEMBER WHO OWNED IT. THE FOUR BACKBONES SHARE THE SAME INPUT AND EACH WAS RUN TWICE, CLEAN AND WITH MASKING.

Backbone	Family	Pretraining	Owner	Protocol and status
MobileNetV2 [32]	CNN	ImageNet	Asif Mahbub	corrected; complete
ViT-B/16 [28]	Transformer	ImageNet-21k	Barshon Basak	corrected; complete
Swin-T [29]	Transformer	ImageNet	Farhana Rahman	corrected; complete
DeiT-S [30]	Transformer	ImageNet	Sami Uddin	corrected; complete
AST [12]	Transformer	ImageNet + AudioSet	Asif Mahbub, Barshon Basak	pre-audit; score NOT RECOVERABLE
2D-CNN, 4 blocks	CNN	from scratch	Asif Mahbub	alternative partition; reference
MobileNetV3-Small	CNN	ImageNet	Asif Mahbub	alternative partition; reference
MobileNetV2 ensemble, $N=5$	CNN	ImageNet	Sami Uddin	calibration carve-out; extension

The Audio Spectrogram Transformer is the only audio-domain pretrained backbone in the project and the only one of the four transformers run before the evaluation audit. It committed no raw confusion matrix, so its challenge-metric score cannot be recomputed. Section III-B reports what does exist and treats the gap as evidence rather than as an omission.

D. Evaluation Metrics

The three abnormal classes are pooled against Normal in the ICBHI 2017 challenge metric. The 4×4 confusion matrix is denoted M . The true class is represented by rows. Normal is Index 0.

$$S_e = \frac{\sum_{i=1}^3 M_{ii}}{\sum_{i=1}^3 \sum_{j=0}^3 M_{ij}}, \quad S_p = \frac{M_{00}}{\sum_{j=0}^3 M_{0j}}, \quad \text{ICBHI} = \frac{S_e + S_p}{2}. \quad (7)$$

A second definition circulates. It is the one our pipeline used before the audit. This one is a macro average of per-class recall and per-class specificity:

$$\text{ICBHI}_{\text{macro}} = \frac{1}{2} \left(\frac{1}{C} \sum_{c=0}^{C-1} \text{Recall}_c + \frac{1}{C} \sum_{c=0}^{C-1} \text{Specificity}_c \right). \quad (8)$$

The two are not close. Each per-class specificity in that macro formula counts true negatives drawn from the other three classes. A rare class, such as Both, scores near 0.95 almost regardless of whether the model ever predicts it. Alongside the challenge metric, we report accuracy, macro precision, macro recall, macro F1, parameters, size, and training time for every model. Confidence intervals resample patients, not cycles, 2,000 times. Two configurations are compared by paired patient-level bootstrap. Where per-cycle predictions exist for both, they are also compared by McNemar's test. The concept-validity gate of Section III-I was fixed before it was run. A concept passes if:

$$\text{AUROC} \geq 0.65 \quad \text{and} \quad \text{the 95 percent DeLong interval excludes 0.5}, \quad (9)$$

The interval is computed by the DeLong method [34]. The 0.65 floor is part of the rule for a reason. An earlier version required only that the interval exclude chance. At $n \approx 2,600$, that is satisfied at an AUROC of 0.52: significant, and useless. A gate that cannot fail is not a gate.

III. MAIN RESULTS

All the runs and their partitions are listed in Table III, together with their score for the challenge metric of (7). The seed replicates, the twelve ablation rows and two concept-gate runs are not models, but variations of the reference configuration. These are reported in Sections III-F, III-G and III-I.

The maximum score throughout this study (from Table III, top-to-bottom) is 0.6864 for the physics-informed auxiliary loss. The next two are 0.6753 for low rank adaptation and 0.6719 for the repeat of the physics-informed run. None of the three is our main result. All are on the 70/30 split where the test set is different, and Section III-H provides the remaining reason.

TABLE III
EVERY MODEL RUN COMMITTED IN THIS PROJECT, WITH THE PARTITION EACH ONE WAS EVALUATED ON.

Run	Partition	Macro	Official	Infl.	S_e	S_p
<i>Main comparison: four backbones, each run clean and with spectrogram masking; detailed in Table IV</i>						
MobileNetV2 + SpecAugment	corrected	0.6140	0.5602	+0.0538	0.4089	0.7115
Swin-T, no masking	corrected	—	0.5304	—	0.3429	0.7179
Swin-T + SpecAugment	corrected	—	0.5291	—	0.3569	0.7013
MobileNetV2, no masking	corrected	0.6075	0.5200	+0.0875	0.4266	0.6135
DeiT-S, no masking	corrected	—	0.5149	—	0.2946	0.7353
ViT-B/16 + SpecAugment	corrected	—	0.5000	—	0.0000	1.0000
ViT-B/16, no masking	corrected	—	0.4994	—	0.0000	0.9987
DeiT-S + SpecAugment	corrected	—	0.4981	—	0.2974	0.6987
<i>Alternative convolutional models</i>						
MobileNetV3-Small	alternative	0.5639	0.5132	+0.0507	0.1980	0.8284
2D-CNN from scratch	alternative	0.5480	0.4720	+0.0760	0.2005	0.7435
<i>Audio-pretrained transformer, trained before the audit; Section III-B</i>						
AST	fallback	0.6359	NOT RECOVERABLE	—	NOT RECOVERABLE	NOT RECOVERABLE
<i>Pre-audit backbone line, kept only to price the evaluation faults</i>						
MobileNetV2 + SpecAugment, pre-audit	fallback	0.7077	0.6495	+0.0582	0.5696	0.7294
Provisional CNN	60/40	0.7181	0.6143	+0.1038	0.3502	0.8784
Tuned CNN, backbone selection	fallback	0.7227	0.6138	+0.1089	0.6118	0.6157
Gated two-backbone fusion	fallback	0.6777	0.5975	+0.0802	0.4852	0.7098
MobileNetV2, pre-audit	fallback	0.6984	0.5895	+0.1089	0.5359	0.6431
<i>Extension experiments; Section III-H</i>						
Physics-informed auxiliary loss	70/30	0.7839	0.6864	+0.0975	0.6980	0.6747
Low-rank adaptation [35]	70/30	0.7969	0.6753	+0.1216	0.7517	0.5989
Physics-informed loss, repeat	70/30	0.7918	0.6719	+0.1199	0.7401	0.6036
Temporal transformer over cycles	70/30	0.6806	0.5832	+0.0974	0.6190	0.5473
Curriculum learning	70/30	0.7046	0.5754	+0.1292	0.6340	0.5168
Dynamic multi-task loss weighting	70/30	0.6377	0.5535	+0.0842	0.5456	0.5614
Multistage distillation [19]	70/30	0.5137	0.5052	+0.0085	0.0932	0.9171
Acoustic curriculum	corrected	0.5022	0.4997	+0.0025	0.0186	0.9808
Demographic feature fusion	70/30	0.6547	0.4733	+0.1814	0.5721	0.3745
Temporal transformer, first attempt	70/30	0.5506	0.3330	+0.2176	0.6660	0.0000
MobileNetV2 ensemble, $N = 5$	carve-out	—	NOT RECOVERABLE	—	NOT RECOVERABLE	NOT RECOVERABLE
<i>Inflation over 20 scored runs</i>		mean +0.0948 , max +0.2176, min +0.0025				

The pre-audit block acts similarly with 0.6495 for the same convolutional model that, after the evaluation is corrected, scores 0.5602.

We first calculate the highest challenge score for the corrected patient independent partition, from the runs with the committed confusion matrix as our main result. Using that rule on Table III will result in the selection of MobileNetV2 and SpecAugment at 0.5602. Table IV is the primary comparison under that protocol, and it provides the complete set of metrics for all four backbones. Each line of its top block was evaluated with the same 2,636 test cycles (from 47 patients) and the same training preprocessing chain and schedule.

TABLE IV
PERFORMANCE OF THE FOUR BACKBONES ON THE CORRECTED PARTITION, 2,636 TEST CYCLES FROM 47 PATIENTS, WITH AND WITHOUT SPECTROGRAM MASKING.

Backbone	Aug.	Acc.	Prec.	Rec.	F1	S_e	S_p	ICBHI [95% CI]
MobileNetV2	yes	0.5880	0.4322	0.4103	0.4141	0.4089	0.7115	0.5602 [0.508, 0.614]
MobileNetV2	no	0.5372	0.3917	0.4091	0.3985	0.4266	0.6135	0.5200 [0.463, 0.575]
Swin-T	no	0.5649	0.3935	0.3826	0.3820	0.3429	0.7179	0.5304 [0.472, 0.588]
Swin-T	yes	0.5607	0.4130	0.4090	0.4058	0.3569	0.7013	0.5291 [0.477, 0.581]
DeiT-S	no	0.5554	0.3483	0.3417	0.3422	0.2946	0.7353	0.5149 [0.466, 0.566]
DeiT-S	yes	0.5349	0.3638	0.3544	0.3571	0.2974	0.6987	0.4981 [0.445, 0.549]
ViT-B/16	no	0.5910	0.1479	0.2497	0.1857	0.0000	0.9987	0.4994 [0.498, 0.500]
ViT-B/16	yes	0.5918	0.1480	0.2500	0.1859	0.0000	1.0000	0.5000 [0.500, 0.500]
<i>Always-Normal baseline</i>		0.5918	0.1479	0.2500	0.1859	0.0000	1.0000	0.5000
<i>Pre-audit protocol: identifier-fallback partition, 492 test cycles from 11 patients, not comparable</i>								
AST	no	0.5528	0.5347	0.4385	0.4109	NOT RECOVERABLE	NOT RECOVERABLE	NOT RECOVERABLE

Table IV shows that MobileNetV2 with SpecAugment is the best model in the table with an ICBHI score of 0.5602 [0.508, 0.614], followed by Swin-T second at 0.5304, and fourth, with the same convolutional backbone, but without masking, was at

0.5200. In detail, it is analysed in Section III-E, and it is compared to published systems in Section III-M.

The always-Normal row is a constant predictor that is labelling all cycles Normal. It achieves a degree of accuracy of 0.5918 which is equal to or above every trained row in the table at a 0.5000 challenge score. We add it since precision on a corpus that is 59 percent Normal is easy to get without learning anything, and the challenge score of (7) separates a trained model from that baseline.

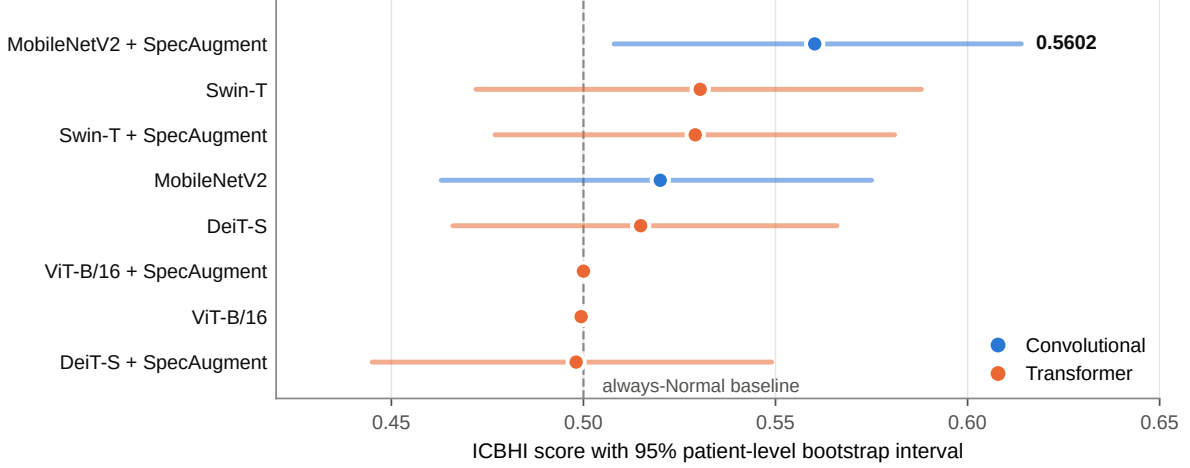


Fig. 2. ICBHI score of the four backbones on the corrected partition, with 95 percent patient-level bootstrap intervals. The dashed line is the always-Normal baseline.

Those scores are displayed in Fig. 2, with the intervals for each patient. Both ViT-B/16 runs sit over each other and each interval overlaps. The ordering in Table IV is not an ordering of point estimates but rather is a ranking of the points on the baseline with an interval too small to resample. This test set is capable of certifying this set of separations.

A. Transformer Results

All the transformers trained under the modified protocol were lost in the convolutional baseline. In Table IV Swin-T outperforms DeiT-S and ViT-B/16 with a score of 0.5304. It is scoring 0.4994 with 12 times the parameters of MobileNetV2. This order is similar to that found in the literature reviewed in Section I-A, All the systems above 0.62 are pre-trained on audio, while none of the three transformers we used are pre-trained. The two rows of ViT-B (16) are the same as the always-normal baseline in all the relevant columns: The variance is too low to resample, so the bootstrap interval is extremely narrow: 0.0000. There are 2,636 predictions, and 2,634 of them are in one of the columns of the confusion matrix. At 85.8 M parameters. It is also the largest model trained on the smallest effective sample size (4262 cycles) across both runs. It was reduced to the majority class. A warm-up schedule with a lower backbone was not tested for the learning rate.

The DeiT-S and Swin-T models are close to convolutional models in terms of accuracy, but have much lower accuracy than they do. These results miss the abnormal cycles, while the sensitivities are 0.3429 and 0.2946, respectively, compared with 0.4089 for the Normal class.

B. The Audio Spectrogram Transformer Run

We use three of the four transforms, all of which have been pre-trained on ImageNet. We have also trained an Audio Spectrogram Transformer [12] with AudioSet weights. The lower part of Table IV gives an accuracy of 0.5528, a macro precision of 0.5347, a macro recall of 0.4385, and a macro F1 of 0.4109 on the identifier-fallback partition. It introduced two changes to the preprocessing to safeguard: a full-band 20-8000 Hz mel filterbank is used instead of 50-2000 Hz, and a 512-point FFT is used in place of a 1024-point FFT. It is 24 times larger than the for comparison: 86.39 M parameters and 329.54 MB at the same time. Transformer results were evaluated against a convolutional backbone and, at 86.74 ms per sample, was compared with 2.94 ms on the same Tesla T4, which is about 30 times slower at inference.

C. Model Cost

Table V shows the number of parameters, the size of the checkpoint, the number of epochs, and the training time (wall-clock) for each model.

ViT-B/16 has 38 times as many parameters and 37 times the checkpoint size of MobileNetV2. In fact, there are 12 times more parameters for a score of 0.0298 in Swin-T, while scores below a constant predictor have lower scores. At this level of data, the more parameters there are, the less accurate the results are. Fig. 3 shows the relationship between two orders of magnitude in model size, the smaller of which has been trained for the backbone, is the best of them.

TABLE V
COMPUTATIONAL COST OF EVERY MODEL. THE TWO GPUS DIFFER BY ABOUT A FACTOR OF TWO, SO TIMES COMPARE WITHIN A DEVICE BLOCK AND NOT ACROSS BLOCKS.

Backbone	Aug.	Epochs run	Best ep.	Time (min)	s/epoch	Params (M)	Size (MB)
<i>Tesla T4</i>							
MobileNetV2	yes	38	38	14.9	23.5	2.23	8.74
MobileNetV2	no	27	27	14.8	33.0	2.23	8.74
2D-CNN, scratch	no	60	19	10.2	10.2	0.42	1.62
MobileNetV3-Small	no	40	17	7.2	10.8	0.93	3.67
Seed replicates ($\times 3$)	yes	40	21–25	17.5	26.2	2.23	8.74
Ablation rows ($\times 3$)	yes	40	12–28	17.8–18.4	26.7–27.7	2.23	9.05
AST, pre-audit	no		not recorded			86.39	329.54
<i>RTX 4050 Laptop</i>							
ViT-B/16	no / yes	40	1	36.2 / 36.3	47.5	85.80	327.31
Swin-T	no / yes	40	18 / 31	20.9 / 20.9	26.2	27.52	104.99
DeiT-S	no / yes	40	2 / 27	11.2 / 11.2	14.1	21.67	82.65
Ablation rows ($\times 7$)	yes	40	12–36	7.5–15.2	11.3–22.8	2.23	9.05

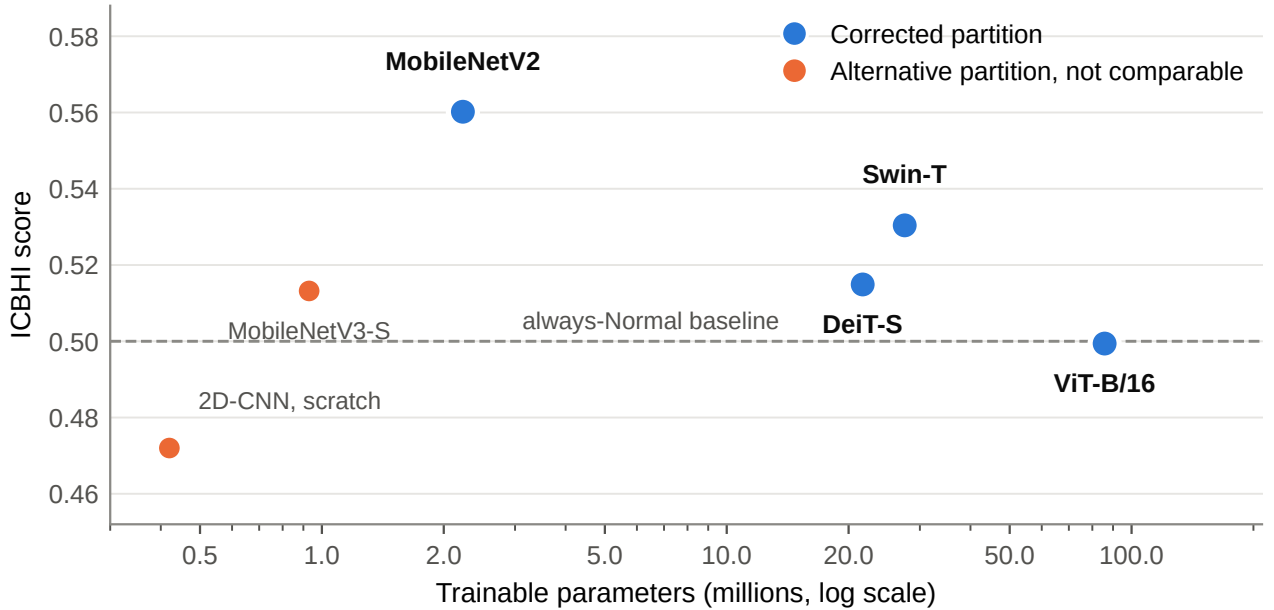


Fig. 3. ICBHI score against trainable parameter count on a log scale. The dashed line is the always-Normal baseline.

The number of parameters does not determine training time either. Within the laptop GPU block, MobileNetV2 takes 21-23s per epoch, while DeiT-S takes 14.1 s with 10 $\times$ the parameters. An increase in the number of parameters that can be saved without compromising GPU performance can be achieved using depthwise-separable convolutions. Only ViT-B/16 is slow in both its senses, with 47.5 s per epoch. This is why we are reporting the number of parameters, the size of the model record, and the training time in separate columns.

D. Data Augmentation

Each backbone was trained twice for each of the two seed types: spectrogram or basic masking was applied to the training set. The deltas in Table IV are: +0.0402 for MobileNetV2, -0.0013 for Swin-T, -0.0168 for DeiT-S and +0.0006 for ViT-B/16. The MobileNetV2 and DeiT-S achieve run-to-run ranges of 0.0141 in Section III-F that are above 0.0141. The noise floor is only cleared by MobileNetV2 as a gain.

Masking aids the pretrained models in convolution, but does not aid transformers. The gain of +0.0402 The largest preprocessing/augmentation effect is 2.9 times the noise range. All are larger than this work, including backbone adaptation, pretraining, and the checkpoint selection rule. On Swin-T and DeiT-S, the composite score is flat or slightly declines. The composite score is at a level or slightly decreasing in Swin-T and DeiT-S. slightly lower, but the macro F1 goes up, from 0.3820 to 0.4058 and from 0.3422 to 0.3571. These two models benefit from masking, as it distributes their predictions across rare classes, thereby improving their accuracy. per-class balance and reduces sensitivity to the abnormal category from which the composite is built. There were no runs in the majority class on the pair, ViT-B/16, so that pair doesn't tell us anything. We

also trained a five-member deep ensemble of the same backbone with a partition that was a calibrated carve-out. The same augmentation increases accuracy from 0.5183 to 0.5447 and macro F1 from 0.3778 to 0.3901.

E. Performance of the Best Model

The pre-experimental selection rule was the highest challenge score on the MCV2, MobileNetV2 with SpecAugment meets it at 0.5602. Fig. 4 and Table VI are both the result file for that run, which was used to create it.

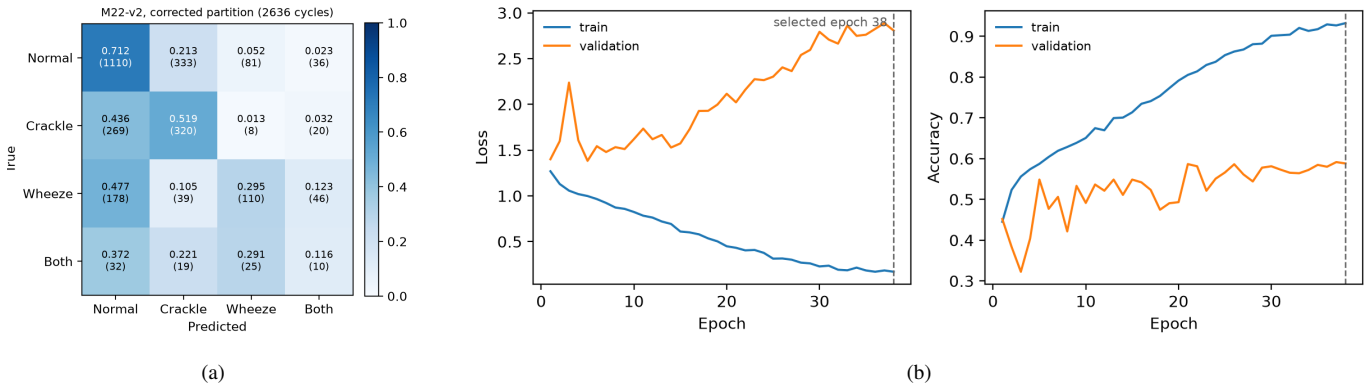


Fig. 4. The best model, MobileNetV2 with SpecAugment. (a) Row-normalised confusion matrix on the 2,636 corrected test cycles. (b) Training and validation loss and accuracy against epoch, with the selected epoch marked.

The confusion matrix is also presented in Fig. 4(b) in row-normalized format. The diagonal mass decreases as the class decreases, with the most abnormal mass being the diagonal mass, which leaks below normal. The diagonal mass is also decreasing as the class decreases, with the heaviest off-diagonal mass corresponding to the abnormal column leaking.

Fig. 4(b) shows the training and validation curves of the same run with the selected epoch marked. Once the monitored score stops improving, the loss continues to drop; this is exactly what it should do. The conflict between the two selection rules costs.

TABLE VI
PER-CLASS PERFORMANCE AND ERROR ANALYSIS OF THE BEST MODEL

Class	Support	Predicted	Precision	Recall	F1	Specificity
Normal	1,560	1,589	0.6986	0.7115	0.7050	0.5548
Crackle	617	711	0.4501	0.5186	0.4819	0.8063
Wheeze	373	224	0.4911	0.2949	0.3685	0.9496
Both	86	112	0.0893	0.1163	0.1010	0.9600
Macro	2,636	2,636	0.4322	0.4103	0.4141	—
<i>Where the errors go</i>						
Abnormal cycles called Normal				479 of 1,076	44.5%	
Abnormal cycles called <i>some</i> abnormal class				597 of 1,076	55.5%	
Abnormal cycles called the <i>right</i> class				440 of 1,076	40.9%	
Normal cycles called abnormal				450 of 1,560	28.8%	
<i>Detection against typing, same predictions</i>						
4-class task			S_e 0.4089, S_p 0.7115		0.5602	
Detect-only task			S_e 0.5548, S_p 0.7115		0.6332	
Cost of sub-typing					0.0730	

Table VI presents the best performance of each class as well as the breakdown of the errors. model. Recall drops from 0.7115 (Normal) to 0.5186 (Crackle) to 0.2949 (Low) on Wheeze and 0.1163 on Both. 597 out of the 1076 abnormal test cycles are marked as abnormal. The abnormal classes are 440 and 479, and the normal class is 440. Out of the 1,560 Normal cycles, 450 are called abnormal.

The bottom block of Table VI rescores the same predictions as a binary detection system. task. The sensitivity increases from 0.4089 to 0.5548 (specificity remains 0.7115), composite goes up from 0.5602 to 0.6332, so the cost of sub-typing is 0.0730. About 40 percent of the remaining errors, the confusion of crackle and wheeze is not one of the causes of failure to detect an abnormal sound at all. This is in line with the human evidence in Section III-K, in which, for the presence of a sound, the physician's agreement is 0.62 and below 0.40 for its finer description [25], [26].

TABLE VII
UN-TO-RUN NOISE FROM THREE SEEDS, AND THE SAME TEN ABLATION ROWS TESTED PER CYCLE BY MCNEMAR AND PER PATIENT BY PAIRED BOOTSTRAP

<i>Noise floor: identical configuration, three seeds</i>				
Seed 42 / 1 / 2		0.5540 / 0.5681 / 0.5657		
Mean, standard deviation, range		0.5626, 0.0075 , 0.0141		
<i>Unit of analysis: the same ten rows tested two ways</i>				
Row	Patient-level 95% CI on the difference	Patient <i>p</i>	Cycle <i>p</i>	Units agree?
C2 random init	[−0.130, +0.003]	0.0625	< 0.001	no
C3 plain CE	[−0.049, +0.023]	0.5625	0.150	yes
C4 frozen backbone	[−0.154, −0.052]	< 0.001	< 0.001	yes
C5 64 mel bins	[−0.061, +0.020]	0.3915	0.026	no
C6 4 s cycles	[−0.092, +0.009]	0.1265	< 0.001	no
P1 + band-pass	[−0.059, +0.034]	0.6860	0.128	yes
P2 + denoising	[−0.089, +0.011]	0.1380	< 0.001	no
P3 + amplitude norm	[−0.017, +0.046]	0.3670	0.040	no
P4 zero padding	[−0.068, +0.006]	0.0955	< 0.001	no
P5 − min-max	[−0.041, +0.027]	0.7170	0.428	yes
<i>Significant at the 0.05 level</i>		1 of 10	7 of 10	6 rows flip

F. Statistical Analysis

As shown in Table VII, the two quantities any ablation delta must be read against are the: run-to-run noise of the training pipeline, and the effect of the unit of analysis on significance testing.

The configuration depicted in the upper section of Table VII was trained three times, the only difference between the runs being the random seed. The scores achieved were 0.5540, 0.5681 and 0.5657, which yielded a mean of 0.5626, a standard deviation of 0.0075, and a range of 0.0141. The reference value employed in our analysis, 0.5602, is within this range. The replicates were obtained by using the ablation harness, while the reference value was obtained from the training notebook. The fact that two separate implementations agree serves as an unplanned reproducibility check. With a noise floor established at 0.0141, three of the eleven ablation deltas are below this level and two are above it by less than a factor of 1.25; the other six results must be treated with caution and are not on their own adequate to support strong conclusions.

The evaluation of the lower block involved two stages of statistical analysis. At the cycle level, McNemar’s test was carried out on 2,636 paired comparisons at the cycle level, which resulted in seven statistically significant and three non-significant outcomes among the ten rows. At the patient level, a bootstrap analysis was conducted on 47 patients; this analysis found only one significant result, the verdict for six of the rows differing between the cycle level and the patient level. Since cycles from the same patient are not independent observations, the effective sample size in the corrected patient-level analysis is 47 instead of 2,636. When combined with the findings from the upper block, this indicates that the majority of the observed differences cannot be accounted for merely by run-to-run noise. The patient-level test set, though, is too small to offer strong statistical evidence for many of the observed ablation deltas.

G. Ablation Study

Table VIII provides the ablation of the best model on the corrected partition. The values of one variable were varied in each row. Each row evaluates or modifies a model component, with 7 rows changing the pre-processing phase. The following rows (P1 to P3) introduce a step that the reference stage does not have. The positive delta in those rows is a recommendation to adopt the configuration that does not have this stage.

Fig. 5 sorts the deltas by the noise band (between rows), such that they carry a claim from the rows that do not. From Table VIII, the greatest single The frozen body of a row is called the effect. C4, which costs 0.1007 and is 7.1 times the noise range. It is also the only row that The patient-level test in Table VII remains important. Removing ImageNet The pretraining price in row C2 is 0.0603. Row C7 does both at once, and gets a score of 0.4979, This is 0.038 higher than the frozen ImageNet backbone of the row C4 and 2.7 times the noise. range. The Frozen ImageNet features have been found to be of inferior quality to random projections on a log-mel input. The movement of the backbone is essential to the payoff of transfer. This divides row C4 into two findings: pretraining is meaningless, and adaptation is needed.

Some rows in the preprocessing stage are left unchanged: in row P1, the cost of band-pass filtering is 0.0089. per-cycle, denoising using spectral-gating in row P2 has a cost of 0.0354. Denoising in row P2 also incurs a cost of 0.0354 when the data is below the noise floor. The value of the normalization amplitude in row P3 is raised by 0.0162. Spectral gating eliminates the stationary Noise and crackles are short transients whose detectability depends on just the low-energy characteristics of the noise. The structure shown in the figure is removed from the gate. Row P3 has the best score in the table: 0.5764; It is not used as a reported model, since its delta is only 1.1 times the noise range. and its patient-level interval is null. Removing the class-weighted loss in row C3 This is below the noise floor (0.0105), and the standard reply to class imbalance is worth considering. The choice of the padding scheme is more important than the choice of the corpus.

TABLE VIII
COMPONENT AND PREPROCESSING ABLATION ON THE BEST MODEL, ONE VARIABLE PER ROW. INTERVALS ARE PATIENT-LEVEL BOOTSTRAP AND THE LAST COLUMN SCALES EACH DELTA BY THE 0.0141 NOISE RANGE

Row	Change from the reference	ICBHI	95% CI	Δ	S_e	S_p	vs noise
REF	MobileNetV2 + SpecAugment	0.5602	[0.508, 0.614]	—	0.4089	0.7115	—
C1	— SpecAugment	0.5200	[0.463, 0.575]	−0.0402	0.4266	0.6135	2.9×
C2	— ImageNet pretraining	0.4999	[0.438, 0.551]	−0.0603	0.3736	0.6263	4.3×
C3	— class-weighted loss	0.5497	[0.502, 0.593]	−0.0105	0.4275	0.6718	<i>below</i>
C4	frozen backbone, head only	0.4595	[0.400, 0.521]	−0.1007	0.4349	0.4840	7.1×
C5	64 mel bins instead of 128	0.5427	[0.492, 0.587]	−0.0175	0.4238	0.6615	1.2×
C6	4 s cycles instead of 8 s	0.5224	[0.463, 0.574]	−0.0378	0.4954	0.5494	2.7×
C7	random init <i>and</i> frozen	0.4979	—	−0.0623	0.0028	0.9929	4.4×
P1	+ band-pass filter	0.5513	[0.492, 0.601]	−0.0089	0.4442	0.6583	<i>below</i>
P2	+ spectral-gating denoising	0.5248	[0.458, 0.588]	−0.0354	0.4572	0.5923	2.5×
P3	+ per-cycle amplitude norm.	0.5764	[0.514, 0.632]	+0.0162	0.3996	0.7532	1.1×
P4	zero padding instead of tiling	0.5291	[0.478, 0.581]	−0.0311	0.4684	0.5897	2.2×
P5	— min-max rescale	0.5539	[0.501, 0.604]	−0.0063	0.4136	0.6942	<i>below</i>

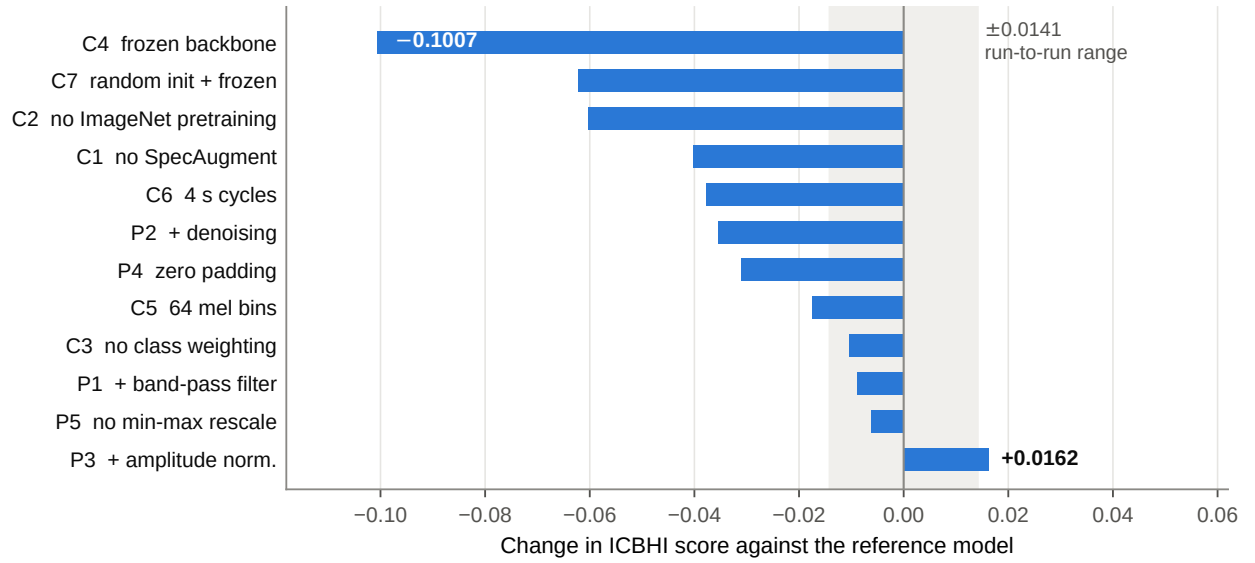


Fig. 5. Ablation deltas against the reference model. The shaded band is the measured run-to-run range of 0.0141.

Table VIII is a leave-one-out ablation: each row contains all of the pipeline. One part of it is removed. Another ablation table is also shown as cumulative form. The two agree only when components do not interact; otherwise, they disagree, and baseline and adding one component at a time.

TABLE IX
CUMULATIVE ABLATION ON THE CORRECTED PARTITION. S3 BRANCHES INTO TWO COMPONENT ORDERINGS AND REJOINS AT S4; THE FINAL COLUMN EXPRESSES EACH DELTA RELATIVE TO THE MEASURED NOISE RANGE (0.0141).

Rung	Added at this rung	Pre.	F.t.	C.w.	Mask	Amp.	ICBHI	Δ vs prev.	vs noise
S0	baseline, random init, frozen, plain CE	×	×	×	×	×	0.5040	—	—
S1	+ ImageNet pretraining	✓	×	×	×	×	0.4911	−0.0129	<i>below</i>
S2	+ full fine-tuning	✓	✓	×	×	×	0.5478	+0.0567	4.0×
S3a	+ spectrogram masking	✓	✓	×	✓	×	0.5497	+0.0019	<i>below</i>
S3b	+ class weighting	✓	✓	✓	×	×	0.5200	−0.0278	2.0×
S4	the remaining one of the two	✓	✓	✓	✓	×	0.5602	+0.0105 / +0.0402	<i>below</i> / 2.9×
S5	+ per-cycle amplitude norm.	✓	✓	✓	✓	✓	0.5764	+0.0162	1.1×

Pre. is ImageNet pretraining, F.t. full fine-tuning, C.w. the inverse-frequency class-weighted loss, Mask spectrogram masking and Amp. per-cycle amplitude normalisation.

The finding is the ordering effect and is greater than the components it reorders. The following are used for masking in the S3a format: credit (S3a): +0.0019, class weight: +0.0105 through S3b class weighting is credited −0.0278 and masking +0.0402. The same two components and the same four runs, with the value of each one running only changing due to the

addition of that one run, by 0.0383. Which is 2.7 times the noise range and bigger than most rows. Table VIII. Both paths start at the same point S2 and end at the same point S4; This is not an interaction, but noise in the endpoints: the two share an overlapping bandwidth: The second one added gets only what the first leaves. If we subtract both leave-one-out deltas from the reference, we predict S2 as 0.5095, for which The measured value is 0.5478, so that the deltas do not add, and a single-ordering table would have This baseline was 0.038 off-base.

Class weighting added on its own at S3b makes the same component added after model *worse* by 0.0278, twice the noise range. The masking at S4 helps by 0.0105. The "go to" solution for class imbalance is not just small on this corpus, as row C3 proposed, but harmful unless masking is in place now to absorb the specificity it costs. The two steps confirm the above findings: The biggest fine-tuning in the ladder at +0.0567 is four times the noise, fine-tuning at S2 The range of values is similar to that of C4, and the cumulative version of C4 is 0.0129 *below* The same direction as the C4–C7 gap is S0. This step is within the box. noise range, and we do not certify it alone; both operating points agree in sign.

Table X shows the effect of masking one metric at a time in the spectrogram, because opposite movements in sensitivity and specificity can occur and be masked by a change in the composite score, specificity.

TABLE X
METRIC-WISE EFFECT OF SPECTROGRAM MASKING ON THE BEST MODEL. A IS THE CLEAN CONTROL; B ADDS MASKING WITH ALL OTHER SETTINGS, INCLUDING THE SEED, FIXED.

Metric	A: no masking	B: + masking	Progress	Possible reason
Accuracy	0.5372	0.5880	+9.46%	fewer Normal cycles called abnormal
Precision	0.3917	0.4322	+10.34%	masking suppresses over-firing on rare classes
Recall	0.4091	0.4103	+0.29%	essentially unchanged; masking adds no coverage
F1-score	0.3985	0.4141	+3.91%	precision gain with recall flat
Sensitivity	0.4266	0.4089	−4.15%	the model becomes more conservative on abnormal cycles
Specificity	0.6135	0.7115	+15.97%	the whole gain sits here
ICBHI score	0.5200	0.5602	+7.73%	specificity gain outweighs the sensitivity loss

Masking raises specificity from 0.6135 to 0.7115 and lowers sensitivity from 0.4266 to 0.4089, The recall remains unchanged at 0.29 percent, but is within a margin of error of 10.34 percent. It doesn't make the It causes the model to detect more abnormal cycles, and it causes the model to no longer call Normal cycles "abnormal. The challenge is sensitive to both metric weights sensitivity and specificity equally, which reads as 7.73 percent. In a screening application, the more you miss the abnormality, the worse it is for you; in the composite, the more you gain, the better off you are. This is a very costly mistake; the same change would be interpreted as a regression.

H. Extension Experiments

Nine extension experiments were carried out together with the four backbones. Most of these experiments have been completed. Before the split correction was applied, they were assessed using a 70/30 split which had been found during our audit. It should be noted that these results cannot be used for ranking against those in Table IV since they were based on an invalid partition; they are given separately in the extension section of Table III, with each experiment being evaluated within the corresponding partition block. Results from different partition blocks are not directly comparable.

The best score attained anywhere in this study was obtained using the physics-informed auxiliary loss. Low-rank adaptation [35], on the same setup, achieved a result that was only 0.011 less than that of full fine-tuning, having trained just 8,356 of the model's 3.64 M parameters (0.23%). Yet we cannot directly assert either of these results since the 70/30 partition employed a different test set from the one used in the corrected repeat of the physics-informed run, which chose epoch 1 as its best epoch. Consequently, the result of the low-rank adaptation should be understood in the context of its own experimental setup and not regarded as a definite overall winner.

Other experiments, including multistage distillation [19], failed. The gated two-backbone fusion detector was very imbalanced and had a sensitivity of just 0.0932. The first temporal transformer attempted never got scores higher than those of either of its backbones, although it also did not go below them. It incorrectly predicted the Normal class, which caused its specificity to be 0.0000.

We give the open-set arm a separate report since it involved a different baseline against which our own comparisons were made. In keeping with the work of Karim *et al.* [18], we reimplemented OpenMax. The unknown-detection AUROC stated by Karim *et al.* was 0.4516, a value that is below the level of chance. By contrast, a post hoc energy score applied to the frozen backbone, without any further training, yielded an AUROC of 0.6466. The earlier claims are less strong than those backed by the 0.6466 result. Nevertheless, given that there were only 19 unknown patients per interval, the sample size is small. For this reason, all open-set results are presented as "not shown to beat" rather than as improvements, as we believe the evidence with this sample size is not sufficient to support stronger claims.

I. Concept Validity Gate

In order for this study to be interpretable, an intermediate layer based on physical acoustic concepts had to be used, since that was the one employed. The clinician was able to read. We established the validity gate given by (9) and allocated a budget of 2 runs. Before we built the extractors, Run 2 used refurbished extractors with failure-mode targets. The threshold and stopping criterion were set before run 1 and were identified in run 1. Both runs were performed. The device was scored in 47 patients over 2,636 test cycles, of which 703 tested positive for crackles, and 459 tested negative. wheeze-positive.

In run 1, the AUROCs for the crackle score are 0.5506; in run 2, they are 0.5580, with the 95 percent confidence interval lying between 0.5495 and 0.5595. having an interval of [0.533, 0.583], an AUPRC of 0.308, and a prevalence of 0.267. The wheeze score reaches 0.5729 and 0.5340, with a range of [0.505, 0.563], and an AUPRC of 0.190 against a prevalence of 0.174; in run 2, both intervals were excluded by chance, and both point estimates are very far from and under 0.65; the gate fails with respect to both concepts, and we applied the stopping rule, which was pre-registered. Among these two concepts, only nine extracted ones have a ground truth. dictionaries, of which five are ICBHI, but two are solely descriptive.

We made a few mechanical changes between the two runs, and they proved effective in each case. The success mode is the opposite of a failure mode. The firing rate of the crackle detector had dropped to about 10 per second. The wheeze detector changed from being silent to sounding its alarm more often, and between 2 seconds 2 and 2.5 it became more certain to sound the alarm, overcoming a saturation problem. In 99 percent of all cases, half of the wheeze cycles occur. We did not attempt a third. It's a difficult one because we have to go through the gate once more in order to correct the extractors. The test setup is installed.

J. Ceiling Estimate

One possible reason is that the ICBHI cycle labels are too noisy for anyone to read. TThe detector reached 0.65, which indicated that the gate was measuring the ceiling set by the benchmark rather than the weakness of our extractors. weakness of our extractors. We tried it before we put it in the readings, as it was before it ran, it was a logistic probe attached to the 79 trained patients and rated on the same. The four representations were tested 2,636 times, and the results are plotted in Fig. 6(a). The The previous registration for reading was that the ceiling explanation was removed when any arm was over 0.80.

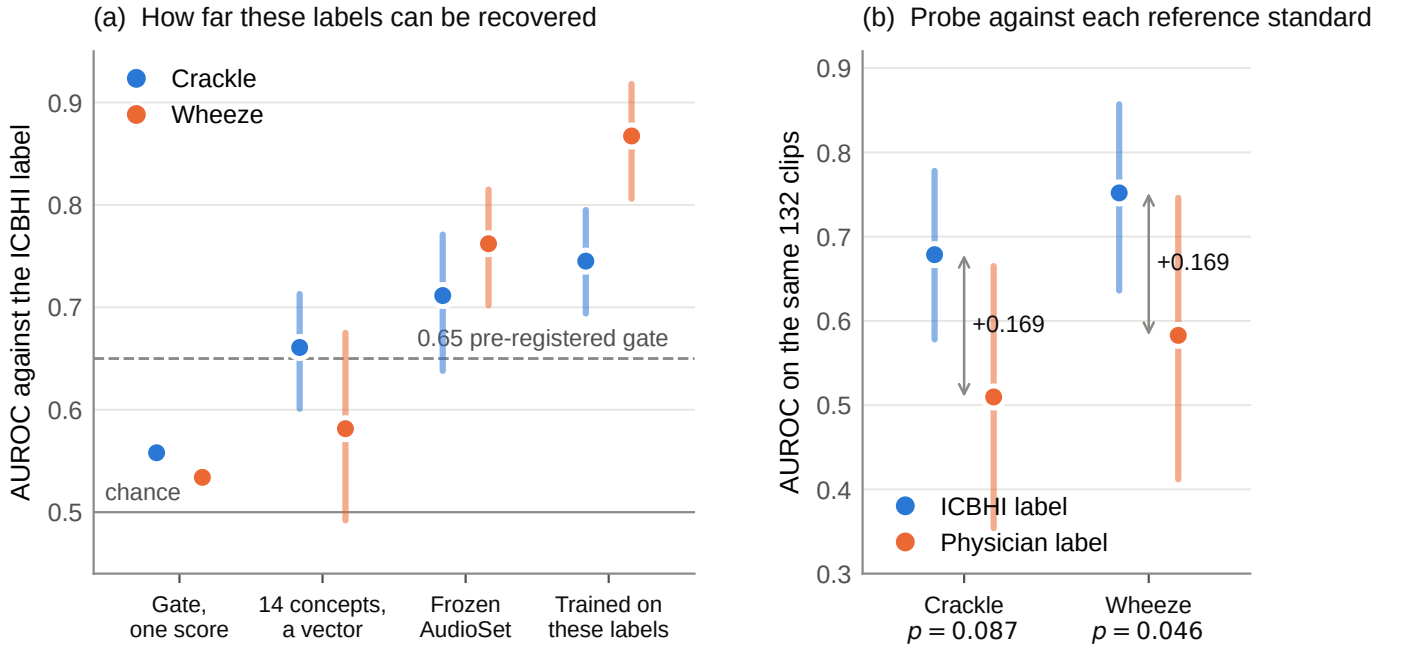


Fig. 6. a) AUROC of four representations against the ICBHI label, with the 0.65 pre-registered gate. (b) The same probe scored against the ICBHI label and against the physician's label on the same clips.

The frozen AudioSet transformer reaches an AUROC of 0.7115 [0.638, 0.771] on crackle and 0.7621 on wheezes. They did not encounter a respiratory corpus or an ICBHI label during the entire period of wheeze. The accuracy of the backbone trained on these labels is 0.7451 and 0.8673. That third arm is where that No one had ever seen an ICBHI label; its performance was not to be taken lightly. It can come to the aid of the trained arm the way it can. We, as a pre-registered branch, fired, so we retract the ceiling claim. These labels help to detect at the level of 0.71-0.87, and it was our own concept that stopped the extractors instead of the reference standards. A flaw in our gate design is revealed through the same comparison. The gate provides a limit on one of the fourteen concepts, taken as a vector, achieve 0.6608 on crackle, compared to hand-built scalar's 0.6051. 0.5815 on wheeze, the crackle figure being above the threshold.

K. Comparison Against a Physician

To report the human agreement with the ICBHI labels, a paired t-test was conducted. A paired t-test was performed to report human agreement with ICBHI labels, as shown in Table XI. A comparison to a machine and a doctor. The upper block collects published studies and our own annotation study, with one physician labeling 132 clips blind with six calibration 12 copies that were not known + exemplars. Upper block scores a frozen AudioSet probe and the All the clips are used by the same physician; the probe is fitted by patient groups using a 5-fold cross-validation So that all clips are out of the sample.

TABLE XI

HUMAN AGREEMENT WITH THE ICBHI LABELS AND A PAIRED MACHINE COMPARISON ON IDENTICAL CLIPS. THE 132 CLIPS WERE NOT PICKED AT RANDOM, SO THE PAIRED DIFFERENCE IS THE RESULT AND NOT THE ABSOLUTE LEVEL.

Study or quantity	Raters / material	Crackle	Wheeze
[24]	7 physicians, 25% of ICBHI test	S_e 23.23%, S_p 72.32%, score 47.77%	
[25]	12 physicians, 20 recordings	κ 0.62	κ 0.59
, fine description	same	$\kappa < 0.40$	
[26]	clinical archive, 11,532 recordings	κ 0.516	κ 0.948
This work , agreement	1 physician, 132 ICBHI clips	κ 0.035 [−0.122, 0.192]	κ 0.266 [0.085, 0.445]
This work , sensitivity	same	23.1% [12.0, 35.3]	29.7% [15.4, 45.2]
This work , intra-rater	12 hidden duplicates	7/8 exact, 11/12 presence	7/8 exact
<i>Same clips, same scores: probe against each reference standard</i>			
Probe against the ICBHI label	$n = 108 / n = 110$	0.6786 [0.578, 0.778]	0.7519 [0.636, 0.857]
Probe against the physician's label	same clips	0.5097 [0.354, 0.665]	0.5828 [0.412, 0.746]
Paired difference	same clips	+0.169 [−0.025, +0.361], $p = 0.087$	+0.169 [+0.002 , +0.344], $p = 0.046$
Physician–ICBHI raw agreement	same clips	0.528	0.718

Our physician moves to the upper block of Table XI and observes that he gets $\kappa = 0.035$ [−0.122, 0.192] on crackle and $\kappa = 0.266$ [0.085, 0.445] on wheeze, at a crackle sensitivity of 23.1 percent [12.0, 35.3]. According to Tzeng et al., sensitivity is reported as 0.83 with an accuracy of 75%. In a separate sample from the same corpus, it was found to be 23.23 percent among seven senior physicians. The Intra-rater duplicates had 7/8 matches on concepts and 11/12 on presence, and inter-rater duplicates had 6/6 matches on concepts and 8/12 on presence. Our rater is self; we have a limit of 3 self-disagreements/12 against a pre-registered limit of 3. consistent. Together, two different measurements that arrive within a "point" of each other say that the low agreement on crackles comes from the material and the labeling convention, and not from one rater.

The lower block score on the probe is 0.6786 [0.578, 0.778] against the ICBHI crackle label. 0.5097 [0.354, 0.665] against the physician's crackle label, and 0.7519 [0.636, 0.857] against The 0.5828 [0.412, 0.746] against the physician's, plotted in Fig. 6(b). Both concepts are paired, with a paired difference of +0.169. significant on wheeze at $p = 0.046$ $p = 0.087$ for crackle remains undecided. On these clips, a network that has never heard a respiratory corpus Is better able to recreate the ICBHI wheeze labels than a self-consistent physician. This says If ICBHI labels mention it, it's not a wrong claim.

L. Explainable Artificial Intelligence

We used the best model to test 400 cycles and applied gradient-weighted class activation mapping [33] and occlusion sensitivity. A heat map alone would be merely decorative; it needs some context, which is why there are panels and the three measurements taken from the panels as seen in Figure 7 and Table XII, are shown in the latter.

TABLE XII

QUANTITATIVE EXPLAINABILITY MEASUREMENTS ON THE BEST MODEL OVER 400 TEST CYCLES. MOBILENETV2 DOWNSAMPLES BY 32, SO EVERY MAP IS A 4 BY 26 GRID INTERPOLATED UP TO 128×801 .

Measure	Condition	Value	Reading
Band pointing	uniform baseline	0.5547	what an unfocused map would give
	wheeze absent ($n = 325$)	0.7674	concentrates in the band
	wheeze present ($n = 75$)	0.7140	concentrates <i>less</i>
	difference	−0.0534	[−0.0785, −0.0256], $p = 0.001$
Tiling consistency	mean correlation ($n = 361$)	0.098	attribution differs across identical audio
	median correlation	0.180	—
	fraction above 0.5	32.1%	only a third are consistent
Padding attention	cyclic tiling, observed / uniform	0.594 / 0.679	ratio 0.87
	zero padding, observed / uniform	0.101 / 0.679	ratio 0.15

This leaves the band-pointing block in Table XII with a focus on the 100 to Groups with and without wheeze: 1000 Hz: 0.7674 and 0.7140, respectively The average is against a base of 0.5547. The difference of −0.0534 [−0.0785, −0.0256] at

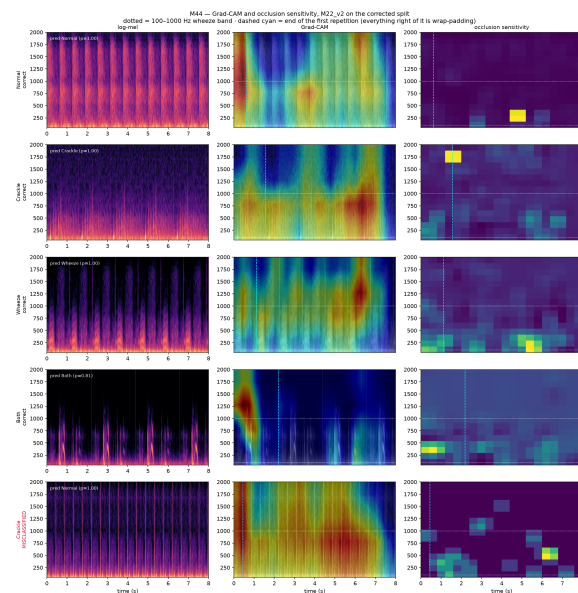


Fig. 7. Attribution panels for the best model. Each row is one test cycle and shows the log-mel spectrogram, the gradient-based attribution, the occlusion map, and the predicted against the true label. The bottom row is a misclassified cycle.

The equation $p = 0.001$ is applicable in the opposite direction to a wheeze detector: the model analyzes the The physiological band is lower when the target sound is present.

The mean correlation value reported by the tiling-consistency block is 0.098, and the median is 0.180. The same repeated copies of one cycle in the same input, the attribution maps of which are identical, are identical with each other. Less than one-third (32.1 percent) of cycles are above 0.5. The modeling of reading acoustics should be made in the same way Multiple copies have been made—one of which follows the position in the window, as a large portion of the attribution is that way. of the sound. The reason for the zero-padding row $\mathbb{P}4$ in this measurement is that we are reading it using the same decoder for the time being. For this measurement, read it using the same decoder as before: Table VIII. The two input schemes exhibit very different behavior in the padding-attention block causing a difference in the padded area of 1.87:1.05. padding. Tilings do carry information, as it costs 0.0311 to replace the tiling with no padding; These measurements are insufficient to determine whether the information is acoustic or positional.

We do not present any of this as clinical validation. The labels of these attributions are aligned. The physician reproduces at $\kappa = 0.035$ on crackles, which is reported in. So, a map aligned to them tells us where the model looks, and not that The model not only reasons the way the clinician reasons, but it is also trained to do so. One planned analysis is not reported, either. The game is a pointing game in which the question is whether the mass lands within the window. The verb to come is not defined in this corpus because ICBHI annotates whether a cycle has an adventitious "to come". This is not sound, and this is not where it happens, and our pipeline crops to this window, so it fixes the Built into a 100 percent hit rate.

M. Comparison with Existing Works

This is compared to published systems on this corpus in Table XIII. Every row is given under the challenge metric of (7), with the partition for each row being named, since this is the argument of the present paper: A comparison without those two columns isn't a comparison. The rows are not rankable relative to each other because they are stored in different blocks. another.

This work is read against the first block and is part of the convolutional systems pre-trained on ImageNet. Each system is pre-trained on the audio below. The four rows above 0.62 are all audio-pretrained. Explain why the ablation of Section III-G predicts the ordering of which. In this study, the most expensive model without the audio-pre-trained backbone is the one that lacks the most. The inclusion of the physician row is due to its being measured on the same labels under the same metric, and A position that is used to put the whole table in context with the reference standard that this corpus is scored against. The trained listeners produced the text, which amounts to 0.4777.

The last block illustrates the importance of the evaluation audit discussed in Section IV: at 0.6495 on a partition that is neither the published one nor patient-independent, the same model would have been placed between two published systems without any changes to the model.

TABLE XIII
COMPARISON WITH PUBLISHED ICBHI 2017 SYSTEMS USING THE CHALLENGE METRIC (7). BLOCKS DENOTE NON-COMPARABLE PARTITIONS;
REPORTED SENSITIVITY, SPECIFICITY, AND CHALLENGE SCORES ARE RETAINED, WITH ACCURACY INCLUDED WHERE AVAILABLE.

Author	Dataset	Network	Acc.	S_e	S_p	ICBHI
<i>Published 60/40 partition, directly comparable</i>						
Gairola <i>et al.</i> [3]	ICBHI 2017	ResNet34, ImageNet	—	0.401	0.723	0.5620
Moummad and Farrugia [14]	ICBHI 2017	CNN6, supervised contrastive	—	0.392	0.760	0.5755
Niizumi <i>et al.</i> [13]	ICBHI 2017	M2D, frozen linear probe	—	0.466	0.722	0.5938
Bae <i>et al.</i> [6]	ICBHI 2017	AST, patch-mix contrastive	—	0.431	0.817	0.6237
Kim <i>et al.</i> [7]	ICBHI 2017	CLAP, text-aided	—	0.457	0.814	0.6354
Jeong <i>et al.</i> [8]	ICBHI 2017	BEATs, patient-aware alignment	—	0.476	0.821	0.6484
Tzeng <i>et al.</i> [24]	ICBHI 2017	seven senior physicians	—	0.232	0.723	0.4777
This work	ICBHI 2017	MobileNetV2 + SpecAugment	—	0.456	0.672	0.5641
This work	ICBHI 2017	MobileNetV3-Small	—	0.198	0.828	0.5132
This work	ICBHI 2017	2D-CNN from scratch	—	0.201	0.744	0.4720
<i>Corrected patient-independent partition, not comparable to the block above</i>						
This work	ICBHI 2017	MobileNetV2 + SpecAugment	0.5880	0.4089	0.7115	0.5602
This work	ICBHI 2017	Swin-T	0.5649	0.3429	0.7179	0.5304
This work	ICBHI 2017	DeiT-S	0.5554	0.2946	0.7353	0.5149
This work	ICBHI 2017	ViT-B/16, collapsed	0.5910	0.0000	0.9987	0.4994
<i>Identifier-fallback partition, shown only to price the fault</i>						
This work	ICBHI 2017	MobileNetV2 + SpecAugment	—	0.5696	0.7294	0.6495

The final block is not a result. It is the same model scored on the 11-patient partition of Fault 2, reported so that the cost of that fault can be read against the published literature: at 0.6495 it would have ranked second in the first block.

IV. CONCLUSIONS

Our goal was to create an interpretable, open-world respiratory sound classifier, but establishing an evaluation protocol proved more difficult. Our concept detectors reached 0.5580 and 0.5340 in two runs using the pre-specified stopping rule for the labels ICBHI crackle and wheeze. This prompted a closer look at the evaluation pipeline, resulting in a fully corrected protocol. According to this protocol, the previously reported score of 0.7077 is reduced to 0.5602, which is 0.1475 more than the effects of the architectural options studied.

The best model we have is MobileNetV2 with spectrogram masking at 0.5602 (corrected partition) / 0.5641 (published partition) with 2.2 M parameters. All three vision transformers failed to reach it, and ViT-B/16 even collapsed towards the majority class, despite having around 38 times as many parameters. The audio-pretrained transformer is not directly comparable, as it predates the audit and lacks a stored prediction table. These ablation results show that fine-tuning makes a significant contribution compared to pretraining; frozen ImageNet features on log-mel input are inferior to fine-tuning; masking mainly improves specificity; and class weighting does not contribute as much as the padding strategy. Several of these conclusions are weakened by the fact that the three-seed noise floor of 0.0141 is quite high, and one conclusion is based on a patient-level test of 47 patients.

The end result of the concept branch was more informative. Interestingly, our detectors performed poorly on the same evaluation cycles, whereas a detector pre-registered with the frozen AudioSet embeddings achieved scores of 0.7115 and 0.7621, indicating that the labels themselves can still be detected and that the problem lay with the extractors we chose. The label produced by the resulting machine system on the same 132 clips was closer to ICBHI wheeze labels than a self-consistent physician (differences 0.169, $p = 0.046$).

The simple recommendation is to report the challenge metric together with sensitivity and specificity, specify the evaluation partition and its patient independence, keep the original confusion matrix, explain the selection rule for checkpoints, and assess statistical claims at the patient level. These practices ensure that results are more easily repeatable, understood, and compared across systems.

Three directions follow. A clean validation protocol and a rerun of the fourth transformer would eliminate the current test-partition selection advantage and place all models in the same test environment. Second, a task-specific reference standard (calibrated by multi-rater re-annotation, and with an explicit “fits no named category” option) might be useful for separating model error from label uncertainty. Third, the fact that fine-tuning strategies that preserve useful representations from large audio-pretrained models offer a promising path forward provides a concrete scale for measuring future adaptation strategies: the frozen probe recovers the target labels at 0.71–0.76 without trainable adaptation, offering a concrete basis for developing future strategies.

This work is based on two principles: setting the decision threshold prior to the experiment and testing the evaluation protocol’s validity before drawing architectural conclusions.

APPENDIX A

RUN IDENTIFIER MAP

Models are named by architecture throughout the paper. Each also carries a short internal identifier, which names the result record every reported number was computed from. Table XIV maps one to the other so that each value above can be traced to its source record.

TABLE XIV
INTERNAL RUN IDENTIFIERS, THE NAME USED IN THIS PAPER, AND WHERE EACH IS REPORTED.

Identifier	Name in this paper	Reported in
M22-v2	MobileNetV2 + SpecAugment — the reference and best model	Tables IV, III
M3-v2	MobileNetV2, clean control, ablation row $\subset 1$	Tables IV, VIII
M40, M41, M42	ViT-B/16, Swin-T, DeiT-S	Tables IV, V
M4 / M43	AST pre-audit / its corrected re-run (NOT RUN)	Section III-B
M1, M2, M3, M12, M22	the pre-audit backbone line and its selection study	Table III
M7	MobileNetV2 ensemble, $N = 5$	Section III-D
M6, M29	OpenMax and post-hoc energy open-set baselines	Section III-H
M30–M37	the extension sweep	Table III; Section III-H
M39	physics-derived concept extraction and the gate	Section III-I
M44, M45	interpretability analysis; component and preprocessing ablation	Tables XII, VIII
M48	protocol ablation, seed band, same-clips comparison	Tables XI, VII
N11	supervised ceiling estimate	Section III-J

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